

Spring of Code

Artificial Intelligence

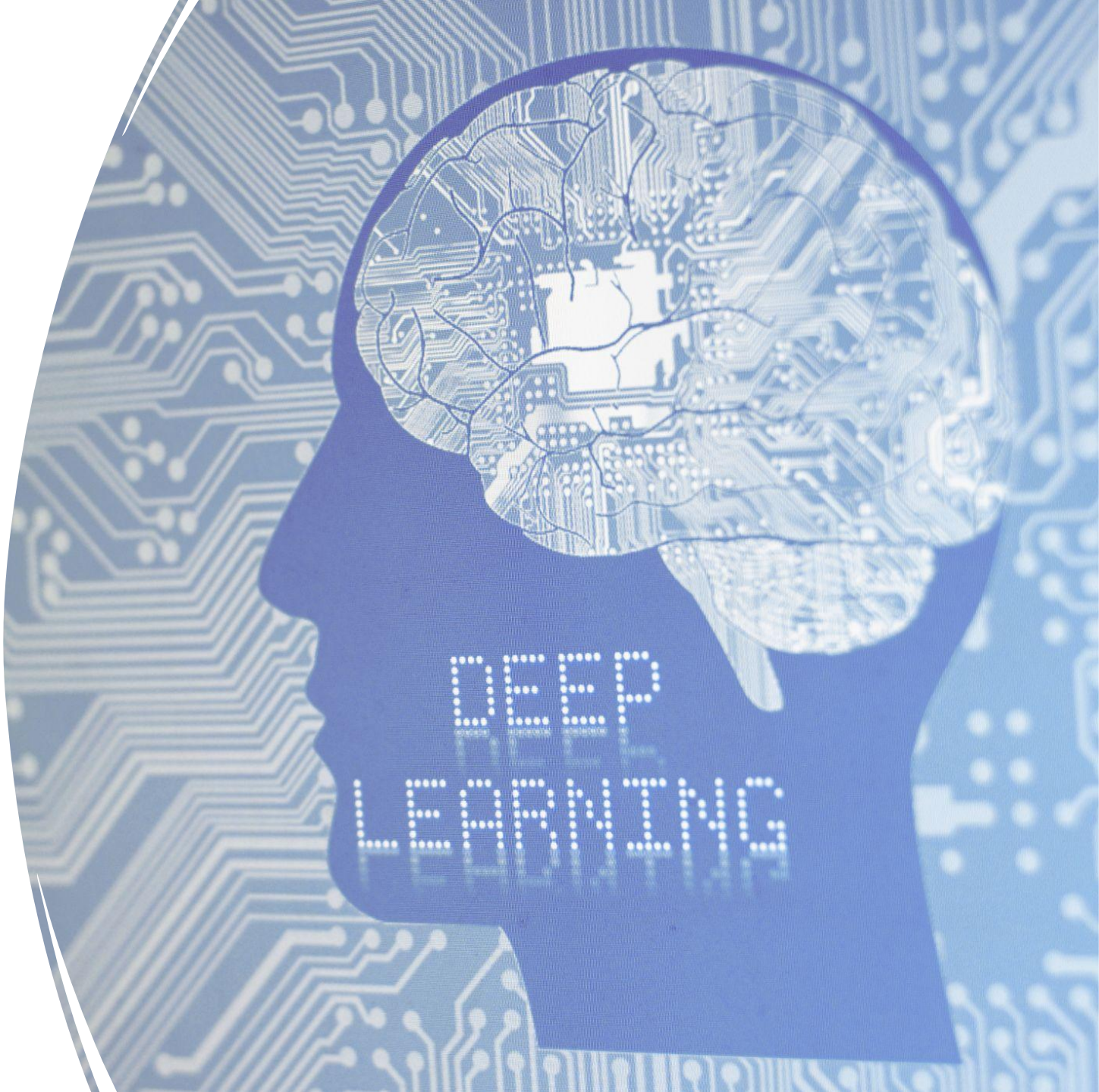
(Machine Learning & Deep Learning)

Instructor
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- *Research Assistant* (DIP Lab)

Duration
03 Months
(Feb – April)

Gated Recurrent Unit (GRU)

- **Objectives:**
  - Introduction to GRU
  - Stacked RNNs
  - Bidirectional RNNs



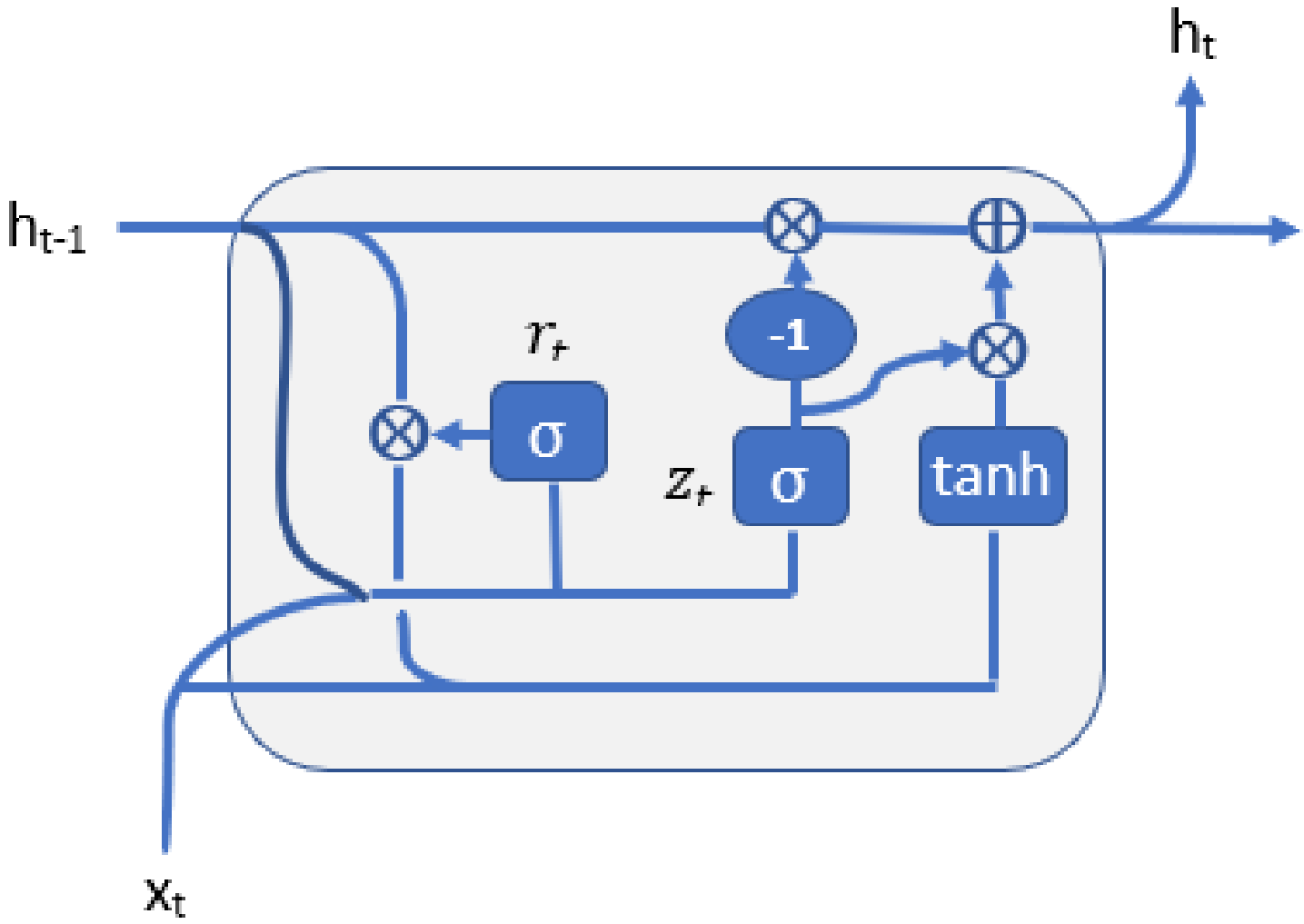


Unlike LSTM, GRU has only one State and only two Gates

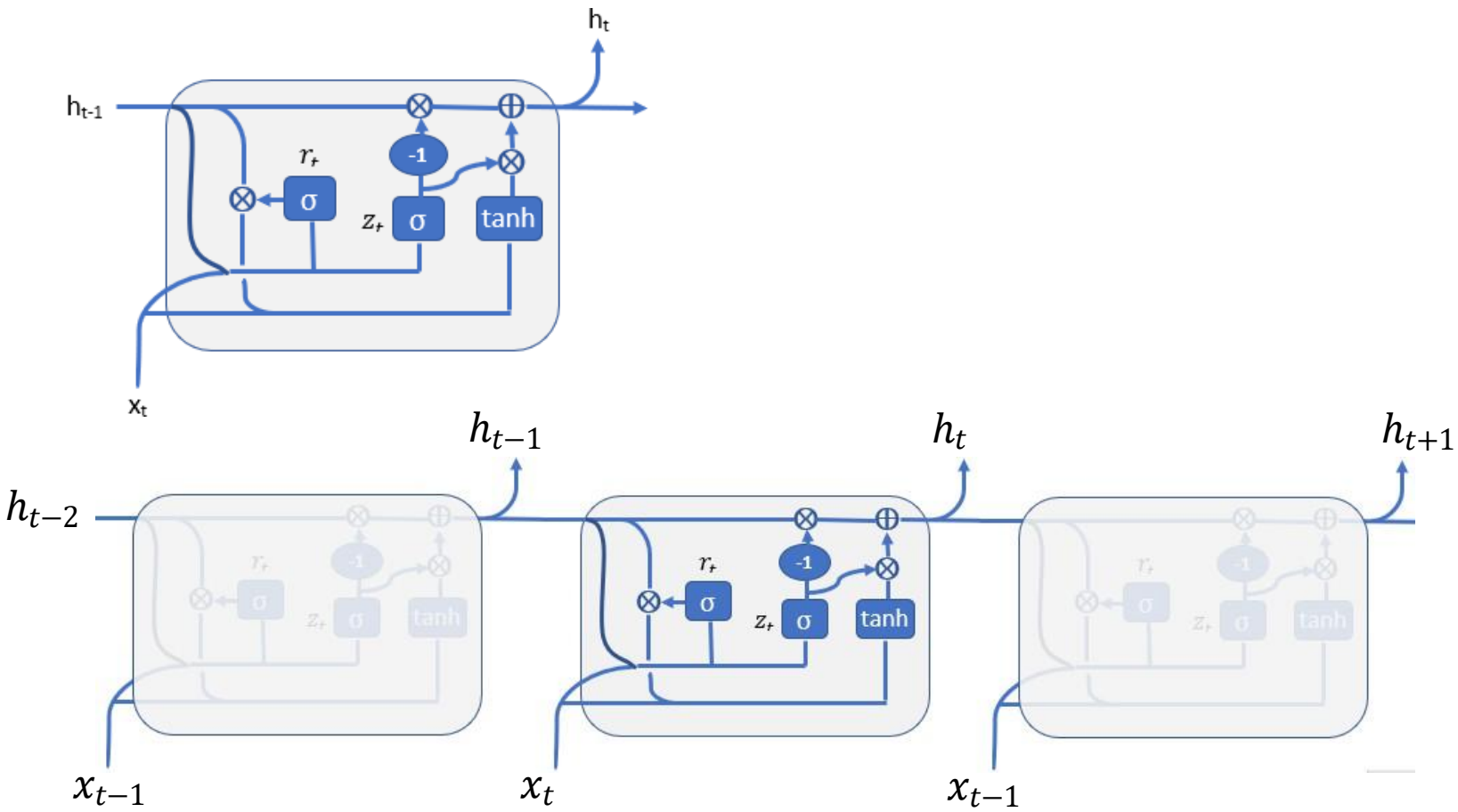


Update Gate and Reset Gate

Gated Recurrent Unit (GRU)

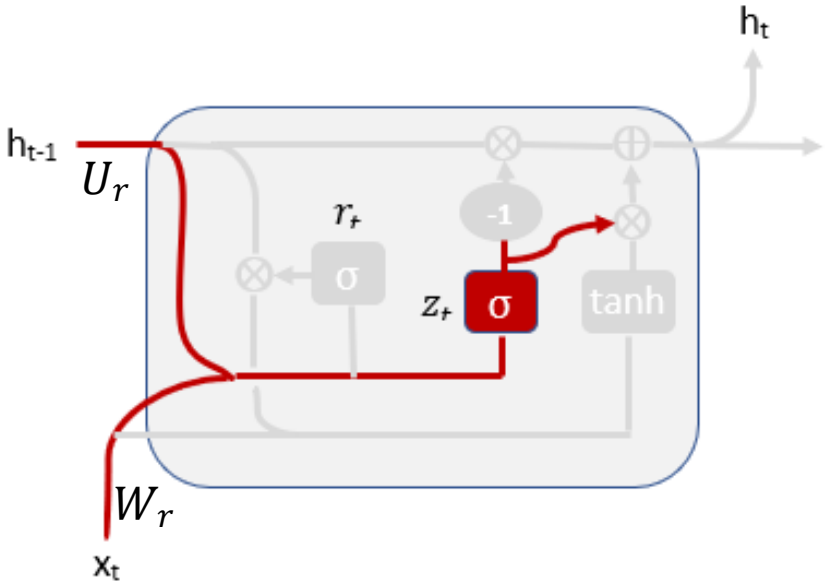


Gated Recurrent Unit (GRU)



Update Gate

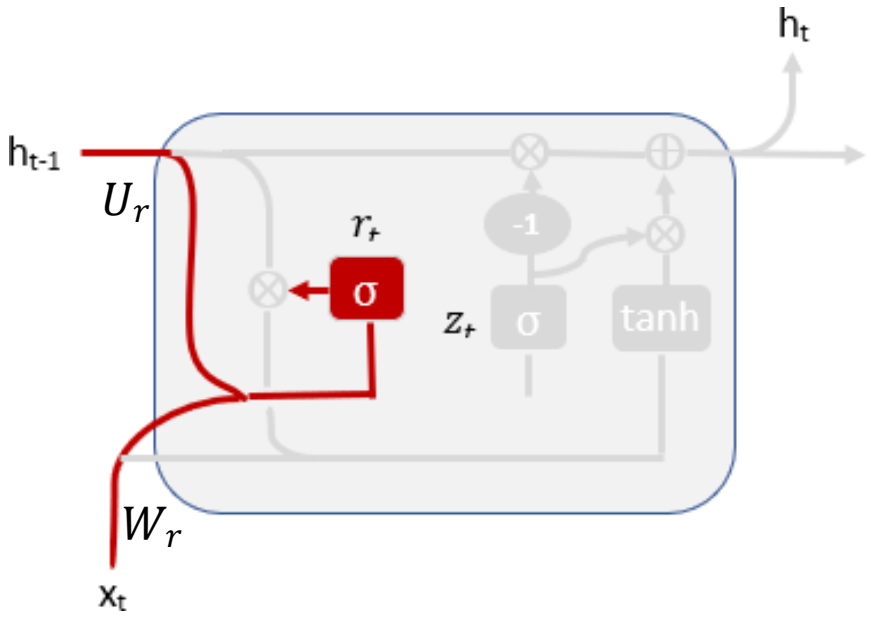
How much of the past information need to be considered to the next time step.



$$Z_t = \sigma(W_z \cdot x_t + U_z \cdot h_{t-1})$$

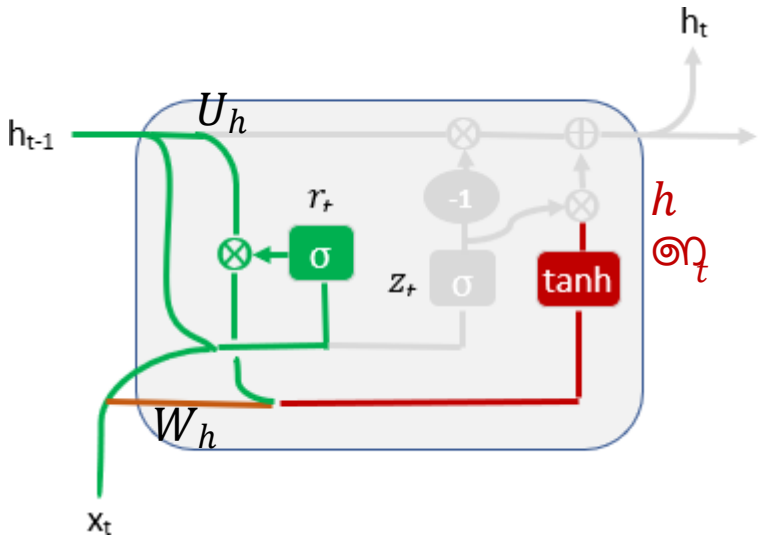
Reset Gate (r_t)

How much of the past information to be forgotten?



$$r_t = \sigma(W_r \cdot x_t + U_r \cdot h_{t-1})$$

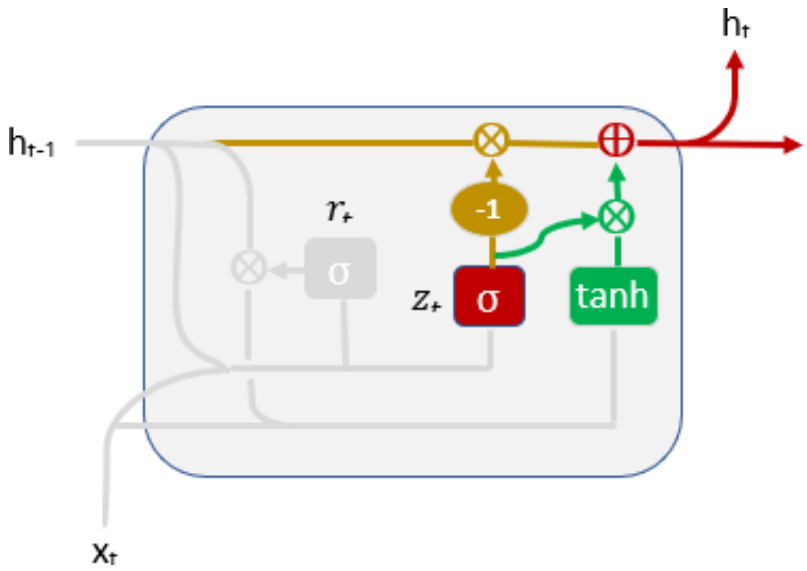
Internal Information (Current Memory Content)



$$h_t = tanh(W_h \cdot x_t + r_t \otimes U_h \cdot h_{t-1})$$

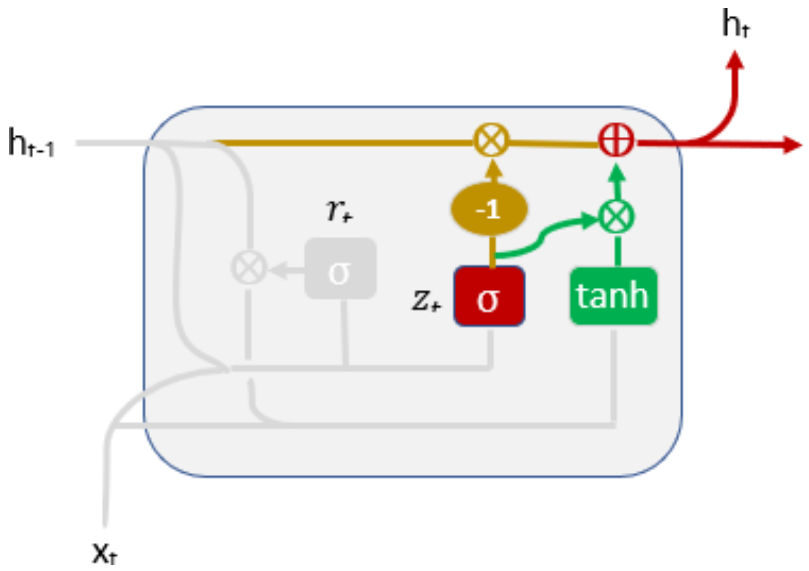
Consider the present input and useful information of the previous hidden state to produce new internal information

New Hidden State (h_t)



$$h_t = (1 - z_t) \otimes h_{t-1} \oplus z_t \otimes \tilde{h}_t$$

New Hidden State (h_t)

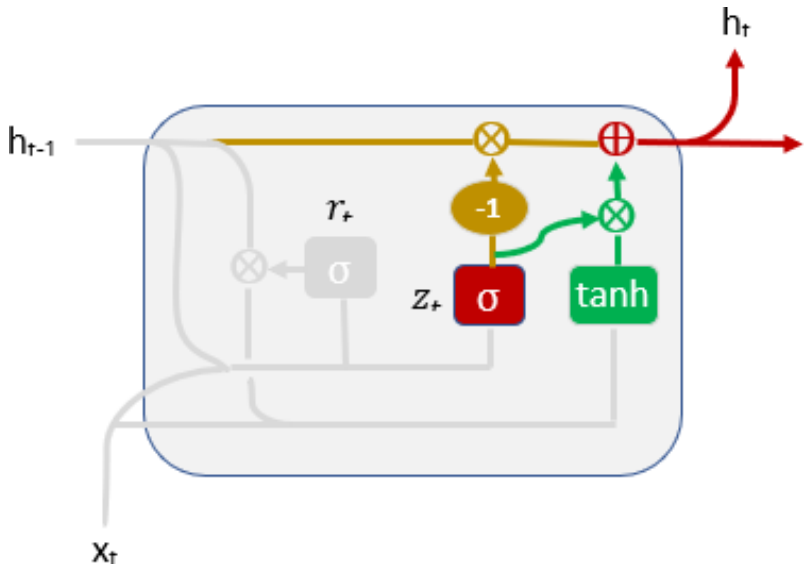


$$h_t = (1 - z_t) \otimes h_{t-1} \oplus z_t \otimes \tilde{h}_t$$

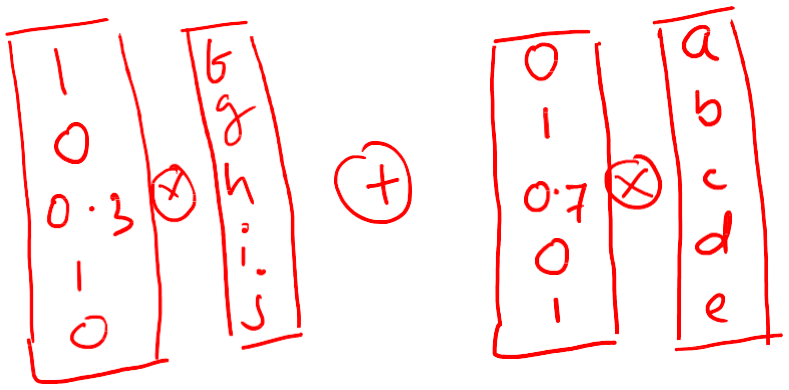
Previous hidden state at the position which are ignored in internal information

Useful information from the internal information

New Hidden State (h_t)



$$h_t = (1 - z_t) \otimes h_{t-1} \oplus z_t \otimes h'_t$$



Summary

One State

- Hidden State $h_t = (1 - z_t) \otimes h_{t-1} \oplus z_t \otimes h'_t$

Two Gates

- Update Gate $Z_t = \sigma(W_z \cdot x_t + U_z \cdot h_{t-1})$
- Reset Gate $r_t = \sigma(W_r \cdot x_t + U_r \cdot h_{t-1})$

Thank You

